LLMs and the Abstraction and Reasoning Corpus: Successes, Failures, and the Importance of Object-based Representations

Yudong Xu *wil.xu@mail.utoronto.ca*
Department of Mechanical & Industrial Engineering, University of Toronto

Wenhao Li *chriswenhao.li@mail.utoronto.ca*
Department of Mechanical & Industrial Engineering, University of Toronto

Pashootan Vaezipoor *pashootan@cs.toronto.edu*
Department of Computer Science, University of Toronto
Vector Institute for Artificial Intelligence

Scott Sanner *ssanner@mie.utoronto.ca*
Department of Mechanical & Industrial Engineering, University of Toronto

Elias B. Khalil *khalil@mie.utoronto.ca*
Department of Mechanical & Industrial Engineering, University of Toronto
Scale AI Research Chair in Data-Driven Algorithms for Modern Supply Chains

Reviewed on OpenReview: *https://openreview.net/forum?id=E8m8oySvPJ*

arXiv:2305.18354v2 [cs.CL] 14 Feb 2024

Abstract

Can a Large Language Model (LLM) solve simple abstract reasoning problems? We explore this broad question through a systematic analysis of GPT on the Abstraction and Reasoning Corpus (ARC) (Chollet, 2019), a representative benchmark of abstract reasoning ability from limited examples in which solutions require some "core knowledge" of concepts such as objects, goal states, counting, and basic geometry. GPT-4 solves only 13/50 of the most straightforward ARC tasks when using textual encodings for their two-dimensional input-output grids. Our failure analysis reveals that GPT-4's capacity to identify objects and reason about them is significantly influenced by the sequential nature of the text that represents an object within a text encoding of a task. To test this hypothesis, we design a new benchmark, the 1D-ARC, which consists of one-dimensional (array-like) tasks that are more conducive to GPT-based reasoning, and where it indeed performs better than on the (2D) ARC. To alleviate this issue, we propose an object-based representation that is obtained through an external tool, resulting in nearly doubling the performance on solved ARC tasks and near-perfect scores on the easier 1D-ARC. Although the state-of-the-art GPT-4 is unable to "reason" perfectly within non-language domains such as the 1D-ARC or a simple ARC subset, our study reveals that the use of object-based representations can significantly improve its reasoning ability. Visualizations, GPT logs, and data are available at `https://khalil-research.github.io/LLM4ARC`.

1 Introduction

It has been recently claimed that Large Language Models (LLMs) such as GPT-4 (OpenAI, 2023a) exhibit "sparks of artificial general intelligence" (Bubeck et al., 2023). As a result, the impressive question-answering and text generation abilities of pre-trained LLMs are already being deployed in rather consequential e-commerce and educational settings[1]. If LLMs are to be used to reliably solve complex, noisy, real-world

[1] `https://openai.com/blog/introducing-chatgpt-and-whisper-apis`

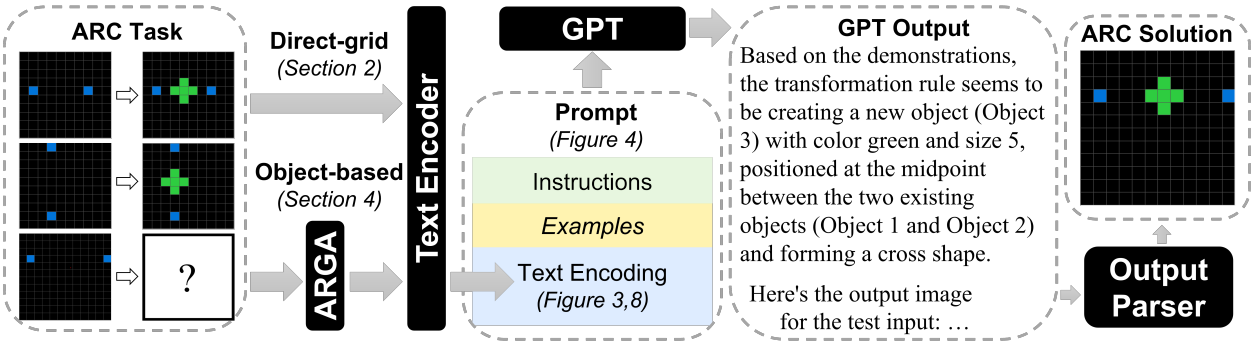

Figure 1: **Example of solving an ARC task with GPT.** An "ARC Task" consists of a set of training input-output pairs followed by a test input for which GPT should produce a correct output. To do so, a prompt is created. It includes high-level instructions on what GPT should do, optionally with additional in-context examples. A text encoding of the ARC task of interest is also included in the prompt. The encoding may be a direct representation of the 2-dimensional grids or an object representation produced by an external ARC solver, ARGA. GPT must then "reason" about the prompt to produce an answer. The output is then parsed and checked for correctness.

problems, one would expect them to be capable of reasoning in simple, unambiguous, idealized settings. By "reasoning", we here mean "using evidence, arguments, and logic to arrive at conclusions or make judgments", as defined in (Huang and Chang, 2022). While the performance of LLMs on arithmetic and language-based commonsense reasoning benchmarks has been the subject of recent analyses (see for example Section 4.1 of (Huang and Chang, 2022) for a brief survey), it is unclear whether LLMs exhibit the ability to generate abstract concepts based on a handful of "training" samples (Odouard and Mitchell, 2022).

To quantitatively measure the gap between machine and human learning, the Abstraction and Reasoning Corpus (ARC) was introduced in (Chollet, 2019). The author advocates for leveraging human-level intelligence as a frame of reference for evaluating general intelligence. To that end, he draws upon the work of developmental psychologists Spelke and Kinzler (2007) on the theory of Core Knowledge to determine axes along which human-like intelligence should be measured. Core Knowledge identifies four broad categories of innate assumptions that form the foundation of human cognition:

- **Objectness:** The ability to perceive the surroundings as consisting of cohesive, persistent, and non-interpenetrating objects.
- **Agentness and goal-directedness:** The tendency to perceive certain objects in the environment as intentional agents with goals, capable of contingent and reciprocal actions, while distinguishing them from inanimate objects.
- **Numerical knowledge:** Innate knowledge of abstract number representations for small numbers and the concepts of addition, subtraction and comparison between those numbers.
- **Elementary geometry and topology:** Knowledge of distance and basic 2D and 3D shapes.

The ARC, a benchmark of 1,000 image-based reasoning tasks, is thus proposed as a test of the above four core knowledge systems in humans or AI systems. Each task requires the production of an output image given a specific input, with 2 to 5 input-output image pairs provided as training instances to "learn" the underlying procedure (Figure 2). The training inputs are different from the actual test input, though they are solvable using the same (unspecified) procedure. Crucially, no acquired knowledge outside of the aforementioned priors is required to solve these tasks. Note that although these priors are explicitly described, the ARC tasks remain completely open-ended: objects can have different shapes and colors and form various relations with one another, and the grid size can also vary between tasks. This feature makes these problems not amenable to solving through search. This is in contrast to games like Go and Chess (Silver et al., 2018; 2017) where

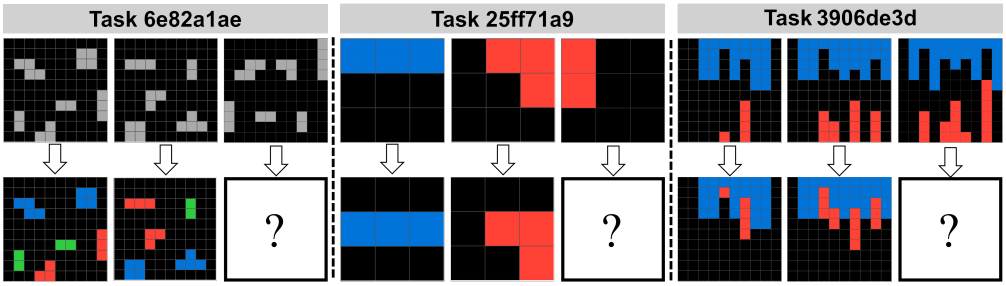

Figure 2: **Sample ARC Tasks.** Three tasks (separated by dashed lines) are shown. For a given task, each column contains one example input-output pair. The first two columns contain the "training" instances and the third column contains the "test" instance. The goal is to use the training instances to solve the test instance. The left task ("Recolor by size") requires recoloring the grey objects to green, red, or blue based on if their size is 2, 3, or 4. The middle task ("Static movement") requires moving the non-black object down 1 pixel. The right task ("Dynamic movement") requires moving the red objects up towards the blue objects until they make contact.

the search space is large but the set of moves is finite and fixed. In fact, so far the approaches that have employed a heuristic search via a predetermined set of transformations have all fallen short of generalizing to the hidden tasks; see Section 5.

Given the vast corpus of human knowledge that LLMs are typically trained on, one might wonder whether they could have acquired the priors listed above. To answer this question, we conducted a comprehensive study of GPT-3.5 and GPT-4 on the ARC. We contribute three high-level findings that we believe shed light on some intrinsic limitations of the LLM framework and, to some extent, how to resolve them:

1. **GPT fails on simple ARC tasks:** Using pure text encodings (Figure 3) of tasks such as those in Figure 2, GPT-4 can only solve 13/50 of the simplest ARC tasks (Section 2). We conduct a failure analysis which reveals that the culprit is the LLM's inability to maintain "object cohesion" across the lines of text that represent the ARC image grids.

2. **GPT does better when objects can be easily detected in text:** We hypothesize that the issue raised in point 1 is tied to the two-dimensional nature of the ARC grids. In Section 3, we introduce the 1D-ARC benchmark, a set of ARC-like tasks that can be represented as a single line of text. Relative to the 50 ARC tasks, GPT performs better on 1D-ARC but is far from perfect.

3. **An object-based representation boosts GPT's performance significantly:** Leveraging the object-centric graph abstractions of the ARC solver Abstract Reasoning with Graph Abstractions (ARGA) (Xu et al., 2023), we provide the LLMs with a more structured object-based representation of the input-output ARC grids (Section 4). This results in a significant jump in performance, where GPT-4 produces the correct output for 23 instead of 13/50 tasks and achieves near-perfect scores on many 1D-ARC task types. Further experiments on additional LLM models such as LLAMA-2 (Touvron et al., 2023) and datasets such as the full ARC training set and the Mini-ARC (Kim et al., 2022), although not the focus of this paper, exhibit a similar pattern.

Given the increasing interest from the artificial intelligence community in the reasoning capabilities of pre-trained LLMs and the unique characteristics of the ARC (and 1D-ARC), we believe that our work contributes to research on imbuing LLMs with such capabilities. We demonstrate that the use of an external tool that produces appropriate representations is crucial. We hope that our experimental design on the ARC, the new 1D-ARC dataset, and the integration of a domain-specific external tool for improved representation will be useful in generating new ideas at the intersection of LLMs and reasoning.